Quiz II

Test your knowledge of different systems that you have learned in this course by completing this quiz.

Quiz ?

Statement

In Google File System (GFS), when a client needs to access data, it contacts the master and gets the data from the master.

Select Multiple Options

- [ ] True
- [ ] False

Check Answer

Quiz ?

Statement

The main responsibility of the GFS master is:

Select Multiple Options

- [ ] To coordinate all read and write requests from clients.
- [ ] To balance the storage load between different ChunkServers.

- [ ] To coordinate concurrent write requests to a file from multiple clients.
- [ ] Store all the filesystem metadata, like file names, total parts, locations of the parts, etc.

Check Answer

Quiz ?

Statement

When scaling systems, mostly distributed system give up:

Select Multiple Options

- [ ] Availability in favor of Partition-tolerance
- [ ] Availability in favor of Consistency
- [ ] Atomicity in favor of Availability
- [ ] Data consistency in favor of availability

Check Answer

Quiz ?

Statement

The motivation for an eventual consistency model was:

Select Multiple Options

- [ ] It is impossible to have highly available replicated data that is fully consistent in a system that can survive network partitioning.

- [ ] Data inconsistencies are highly undesirable, so the primary focus should be on ensuring that all replicas are consistent.
- [ ] Distributed commit protocols can never work reliably, so data is bound to become inconsistent on some participants.
- [ ] Data might be in an inconsistent state during the execution of transactions but will be made consistent when they commit.

Check Answer

Quiz ?

Statement

In BigTable, a Tablet refers to:

Select Multiple Options

- [ ] A subset of rows and column families
- [ ] A contiguous range of rows of a table
- [ ] All columns of contiguous rows of a Locality Group
- [ ] All columns families of contiguous rows of a table

Check Answer

Quiz ?

Statement

Split-brain is a condition that arises when:

Select Multiple Options

- [ ] A leader election is taking place.
- [ ] A network is partitioned.
- [ ] The election algorithm fails to elect a leader.
- [ ] The leader initiates a leader election.

Check Answer

Quiz ?

Statement

The main difference between the Google File System (GFS) and the Hadoop Distributed File System (HDFS) is that HDFS:

Select Multiple Options

- [ ] HDFS stores all filesystem metadata on a single master.
- [ ] HDFS stores files in fixed-size blocks.
- [ ] HDFS does not allow concurrent reading of a file.
- [ ] In HDFS, there are no concurrent writers and arbitrary file modifications.

Check Answer

Quiz ?

Statement

Distributed systems use the following techniques for Fencing. Select all that apply.

Select Multiple Options

- [ ] Resource Fencing
- [ ] Network Fencing
- [ ] Node Fencing
- [ ] None of the above

Check Answer

Quiz ?

Statement

In distributed systems, Fencing is used to:

Select Multiple Options

- [ ] To isolate a central server from other servers.
- [ ] To isolate a server while it initializes.
- [ ] Isolate a faulty server to prevent it from doing any damage or causing corruption.
- [ ] To enhance security around certain critical nodes.

Check Answer

Quiz ?

Statement

A column family in BigTable is:

Select Multiple Options

- [ ] A group of columns within a single table.
- [ ] A group of related columns among multiple tables.
- [ ] A set of columns that are shared across multiple tables.
- [ ] A preconfigured set of columns to make it easier to set up a Bigtable instance.

Check Answer

Quiz ?

Statement

Dynamo uses vector clocks to:

Select Multiple Options

- [ ] To garbage collect older versions of an object.
- [ ] To store multiple versions of data.
- [ ] Determine causality between different updates on the same object.
- [ ] To serialize concurrent updates to an object.

Check Answer

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Quiz ?

Statement

In distributed systems, replication gives following benefits. Select all that apply.

Select Multiple Options

- [ ] Availability
- [ ] Consistency
- [ ] Reliability
- [ ] Better read performance

Check Answer

Quiz ?

Statement

In distributed systems, Heartbeat is used to:

Select Multiple Options

- [ ] Detect dead servers
- [ ] Synchronize operations between servers
- [ ] Measure load on servers
- [ ] Transfer configuration changes

Check Answer

Quiz ?

Statement

What statements are true about the Google File System (GFS)? Select all that apply.

Select Multiple Options

- [ ] Different parts of a file can be stored on different servers.
- [ ] GFS can store multiple replicas of a file's data on different servers.
- [ ] Any file stored in GFS can be bigger than the capacity of a server.
- [ ] Transfer configuration changes

Check Answer

Quiz ?

Statement

**What is the importance of choosing a good row key in BigTable? **

Select Multiple Options

- [ ] Take advantage of the locality of adjacent rows.
- [ ] Avoid the use of duplicate keys.
- [ ] Enable rapid search for an arbitrary row based on its key.
- [ ] Make queries user-friendly for easier debugging.

Check Answer